# VLB entry changes: go-live plan

Implementation, testing, release and rollback for the findings in the VLB entry procedure review

Caliber Pulse MIS | Companion to docs/vlb-entry-procedure-review-2026-11-02.pdf | Repository main at commit d7486f8 | Prepared 2 November 2026

## 1. What this plan covers

The review found eleven items. This plan turns them into nine code changes grouped into three releases, each small enough to go out on its own and be reverted on its own. Every change is code only except the optional auto-save setting in release 2, which adds one nullable column. No existing batch, card or scan is rewritten by any release.

The live site is Pulse-II at pulse-ll.cmll.in on Azure App Service caliber-mis-783. A push to main runs the GitHub Actions workflow main_caliber-mis-783.yml, which installs, prepares the OCR model, builds, runs the test suite, verifies the release identity and server start, deploys the full workspace, and then polls the live health endpoint until it reports the new commit. A new build changes the application version, so every open browser session is logged out and reloaded within ten seconds of the deployment going live.

### Release grouping

| Release | Changes | Why together | Needs a team decision first |
|---|---|---|---|
| R1 Stop wasted entry work | 1, 2, 3, 4 | All four prevent the user from starting work the server will refuse. No behaviour of saved data changes. | No |
| R2 Scan posting rule | 5 | Whether a clean scan card may post without an Approve click. Code or documentation changes depending on the answer. | Yes, section 3 |
| R3 Form quality | 6, 7, 8, 9 | Meter warnings, transfer hold from the Manual form, Edit and Delete state, shared PDF limits. Larger UI surface, so it ships after R1 has settled. | Partly, meter blocking rule |

## 2. Release 1: stop wasted entry work

### Change 1. Batch picker and entry workspace respect stage and amendment owner

Review finding 1

**Files:** src/vlb-create-batch.jsx (openVlbEntryBatches, view, entry auto-open effect), src/vlb-entry-workspace.jsx, vlb-batches.mjs (batch list fields)

**Behaviour after the change**

- The entry picker lists a batch as openable only when it is not closed, its stage is vlb_open (or two-stage is off), and either it has no amendment owner or the owner is the signed-in user.
- Batches that fail the test still appear, greyed, with the reason: "VLB locked, ask the site PM to unlock", "Reopened for <user>", or "Closed".
- The entry workspace mounts the Manual form or Scan page only for a batch that passes the same test, and unmounts it when the ten-second poll reports the batch has been locked or closed by someone else, showing the reason in place of the form.
- Auto-open of a single matching batch uses the same filter, so a site whose only open batch is in the ELB stage shows the reason instead of opening the form.

**Tests to add or extend**

- test/vlb-batches-ui.test.mjs: picker hides Open for elb_open and foreign-amendment batches and shows the reason text.

• test/vlb-entry-workspace.test.mjs (new): the form unmounts when the polled batch turns elb_open or closed.

**Schema:** None

**Risk if wrong:** A user with a legitimately writable batch sees it greyed. Mitigation: the server rule in assertVlbBatchWrite is unchanged, so the worst case is a blocked screen, never a wrong save.

## Change 2. Scan batch creation checks for an open VLB batch

Review finding 2

**Files:** bulk-card-scanning.mjs (POST /api/scan-batches, requireContextReady), local-document-scanning.mjs (batch helper used by /files, /folder, /capture), scan-batch-context.mjs (gate shape), src/bulk-card-scanning.jsx (gate message)

**Behaviour after the change**

• Creating a vehicle scan batch calls matchingVlbBatch for the expected context and refuses with 409 "No open VLB batch for <site>, <date>, shift <n>. Create or open it on Site batches first." when none exists, when it is closed, or when its stage is elb_open.

• The same check runs in requireContextReady so Save all and auto-save fail early with the same message if the VLB batch was locked after upload.

• The scan page shows the gate message in the Scan setup area before any upload control is enabled.

• Equipment scan batches are not affected.

**Tests to add or extend**

• test/bulk-card-scanning.test.mjs: POST /api/scan-batches returns 409 with no VLB batch, with a closed batch, and with an elb_open batch; 201 with a vlb_open batch.

• test/local-document-scanning.test.mjs: /files and /folder refuse with the same message for vehicle jobs.

**Schema:** None

**Risk if wrong:** Scanner users at a site whose MIS user has not yet created the batch are blocked from uploading. That is the intended order of work and matches what the Manual path already enforces.

## Change 3. Shift lookup falls back on start time and warns before entry

Review finding 4

**Files:** src/vlb-entry-context.mjs (vlbEntryFields), src/log-book-form.jsx (shift field hint), trip-card-rules.mjs (selectedMasterShift, exported already)

**Behaviour after the change**

• vlbEntryFields resolves the Shift Master row through selectedMasterShift, which matches by code, then by start time, and prefers the dated revision in force on the batch date.

• When no row resolves, the form shows a red notice above the fields: "No Shift Master row for shift <code> at <site> on <date>. Add it in Shift master before entering cards." and the Save button is disabled.

• Pulse hours lookups work whenever a row resolves, which they do not today for batches whose shift is only matched by start time.

**Tests to add or extend**

• test/vlb-entry-context.test.mjs (new): resolves by code, by start time, prefers the dated revision, returns empty shiftId with a reason when nothing matches.

• test/log-book-form-ui.test.mjs: Save disabled and notice shown when shiftId is empty in batch mode.

**Schema:** None

**Risk if wrong:** A site with two Shift Master rows sharing a start time but different names stays unresolved, as it does today; the notice now says so.

### Change 4. Approve runs the context gate

Review finding 9

**Files:** bulk-card-scanning.mjs (PUT /api/scan-cards/:id with action approve)

**Behaviour after the change**

- Approving a card calls requireContextReady, so approval is refused while any card in the batch is still reading or disagrees on date, site or shift. The message is the existing gate text.
- Save and reject are unchanged.

**Tests to add or extend**

- test/scan-correction-status.test.mjs or test/scan-batch-context.test.mjs: approve returns 409 while the gate is blocked, 200 once resolved.

**Schema:** None

**Risk if wrong:** Reviewers must resolve mismatched cards before approving any card, which Save all already demands.

## 3. Release 2: scan posting rule

Today a scan card in review status with clean checks posts to the log book automatically. The written procedure in SCAN-REVIEW-GROUPS.md and GEMINI-SCANNING.md says no card posts without approval. The team must choose one of two options before release 2 is built.

### Option A: approval required for every card

### Change 5A. Auto-save requires approved status

Review finding 3

**Files:** scan-auto-save.mjs (canAutoSaveScanCard), src/bulk-card-scanning.jsx (auto-save effect, progress text), SCAN-REVIEW-GROUPS.md

**Behaviour after the change**

- canAutoSaveScanCard returns false unless the card status is approved and checked_by is set. Clean cards in review status wait for the reviewer.
- The Save button on a single card is unchanged; it still requires the card to be approved, which the UI already enforces by hiding it otherwise.
- Progress text "Saving complete card N automatically" is kept for approved cards that save in the background.

**Tests to add or extend**

- test/scan-auto-save.test.mjs: review-status card with clean checks returns false; approved card returns true.

**Documentation:** No change; the documentation already describes this behaviour.

**Schema:** None

**Risk if wrong:** Throughput per batch drops by one click per card. For a 100-card shift that is one reviewer pass the team currently skips.

### Option B: keep auto-save, make it visible and switchable

### Change 5B. Auto-save becomes a per-batch setting with a visible banner

Review finding 3

**Files:** scan-auto-save.mjs, bulk-card-scanning.mjs (scan batch create and detail), src/bulk-card-scanning.jsx (Scan setup toggle, banner), SCAN-REVIEW-GROUPS.md, GEMINI-SCANNING.md

**Behaviour after the change**

- Scan setup gains a switch "Post complete cards automatically", default on to match today. The value is stored on the scan batch and returned with batch detail.
- canAutoSaveScanCard receives the batch setting and returns false when it is off.
- The review page shows a persistent banner while the setting is on: "Complete cards post to the VLB without an Approve click. Cards with warnings still wait for you."

- The documentation is corrected to describe the switch and the default.

**Tests to add or extend**
- test/scan-auto-save.test.mjs: setting off blocks auto-save; test/bulk-card-scanning.test.mjs: setting persists and is returned.

**Schema:** One additive column: ALTER TABLE mis_scan_batches ADD COLUMN IF NOT EXISTS auto_post BOOLEAN NOT NULL DEFAULT TRUE. Existing batches keep today's behaviour.

**Risk if wrong:** None to saved data. Users who prefer approval per card turn the switch off per batch.

# 4. Release 3: form quality

### Change 6. Meter rule breaches shown inline before save

Review finding 5

**Files:** src/log-book-form.jsx (meter fields, totals), trip-card-rules.mjs (meterRuleIssues, already shared), log-book.mjs (optional blocking)

**Behaviour after the change**
- The form evaluates meterRuleIssues on every change and shows each breach beside the field it concerns: closing below opening, working HMR above shift hours, KMR above the limit.
- The total fields show the signed value, not a dash, when negative.
- Decision needed: whether closing below opening blocks the save (server 400) or remains a warning stored in meterRuleErrors as now. The recommendation is to block closing below opening and keep the two limit rules as warnings, because a negative reading is always a transcription error while the limits have genuine exceptions.

**Tests to add or extend**
- test/log-book.test.mjs: 400 for closing below opening when blocking is chosen; meterRuleErrors stored for limit breaches. UI test: breach text rendered next to the field.

**Schema:** None

**Risk if wrong:** If blocking is chosen, a genuine meter reset at a site is refused until corrected; the message must say what to do.

### Change 7. Manual form can raise the awaiting-transfer hold

Review finding 6

**Files:** log-book.mjs (GET /api/log-books/options returns twoStage), src/log-book-form.jsx (validateTripEquipment anySite, equipment dropdown), src/log-book-asset-options.mjs, src/vlb-batch-holds.jsx (already shows holds)

**Behaviour after the change**
- With two-stage on, the loading machine dropdown lists equipment from other sites below a divider, each tagged with its current site.
- Choosing one passes client validation; the server records the site hold as it already does for scan cards, and the batch holds panel offers the express transfer OTP.
- With two-stage off, behaviour is unchanged: site-only.

**Tests to add or extend**
- test/log-book-form-ui.test.mjs: other-site equipment listed and tagged when twoStage is true; absent when false. test/log-book.test.mjs: manual save with other-site loader stores a vlb_loader hold.

**Schema:** None

**Risk if wrong:** Operators may pick a wrong-site machine by mistake. The tag and the hold panel make it visible, and the card is held, not posted as working.

### Change 8. Edit, Delete and navigation follow batch state

Review finding 7

**Files:** src/log-book-form.jsx (Edit, Delete, First, Last, register load)

**Behaviour after the change**

- Edit is disabled with a tooltip reason when the batch stage is not vlb_open, when the batch is closed, or when an amendment owner is someone else.
- The Delete button and its dialog are removed from the batch form; a link "Delete through the Deletion page" replaces them, since direct deletion always returns 403.
- First and Last move within the batch records, not the global register, so a card from another batch can no longer be opened in this form.

**Tests to add or extend**

- test/log-book-form-ui.test.mjs: Edit disabled per state; Delete button absent; First and Last stay within the batch list.

**Schema:** None

**Risk if wrong:** None to data. Users lose a button that never worked.

### Change 9. One shared module for PDF and image upload limits

Review finding 8, 11

**Files:** scan-upload-limits.mjs (new, shared by browser and server), src/bulk-card-scanning.jsx, src/scan-upload-image.mjs, local-document-scanning.mjs (renderSavedPdf, readUploads, /files, /folder), operational-start.mjs import in five UI files

**Behaviour after the change**

- Page limit (120), target render size, image format, per-file and per-upload byte limits and their messages come from one module. The browser and the local scanner path render to the same size and format.
- readUploads hashes each PDF and consults local_scan_files like the folder path, so a re-uploaded PDF is refused before extraction with the existing "belongs to an earlier batch" message.
- The hard-coded 2026-11-01 in the UI is replaced by the operational start constant.
- Housekeeping from finding 11: remove the duplicate assertVlbBatchWrite call in insertLogBook, drop the unused session-storage write, correct the status line and reopen wording in docs/vlb-elb-batch.md.

**Tests to add or extend**

- test/local-document-scanning.test.mjs: duplicate PDF via /files refused; limits read from the shared module. test/scan-image-format.test.mjs: both paths produce the same dimensions for a sample page.

**Documentation:** docs/vlb-elb-batch.md status line and section 3 reopen wording; SCAN-PERFORMANCE.md limits table.

**Schema:** None

**Risk if wrong:** Changing the local render size alters OCR input for the Windows scanner path. Keep the larger of the two sizes, verify extraction on ten real cards before release.

## 5. Build and verification before each release

### 5.1 Branch and pull request

1. Branch from main: fix/vlb-entry-r1, fix/vlb-entry-r2, fix/vlb-entry-r3. One pull request per release.
2. Each change lands as its own commit with the tests in the same commit, so a single change can be reverted without touching the others.
3. Run the suite locally before pushing. On Windows expect the two known environment failures (CRLF in the deletion UI test, release temp path); fix the CRLF test in R1 by making its import-stripping pattern accept \r\n so the suite is green on Windows too.

```
npm ci
npm test
npm run build
```

### 5.2 Continuous integration

ci.yml runs on the pull request: install, test, build, and the browser smoke step. The pull request merges only when it is green. The deploy workflow runs the same test suite again on main, so a red test never reaches Azure.

### 5.3 Local walk-through with a database

Run the server against a local PostgreSQL copy restored from the Backup page export, as ERP-LOCAL.md describes, and walk the scripts in section 6 with an MIS login and a PM login. This is the only step that exercises the real screens; the unit tests cover the rules but not the layout.

```
$env:DATABASE_URL = "postgresql://USER:PASSWORD@127.0.0.1:5432/caliber_mis"
$env:DATABASE_SSL = "false"
npm run build
npm start
```

## 6. Acceptance scripts for the team

Each script is run once on the local copy before merge and once on the live site after deployment, by an MIS user at one site. Record pass or fail per line.

**Script A: Manual entry, release 1**

1. Open Vehicle Log Book, VLB Manual, choose a site with no open batch. Expect: "Create the batch from Create Batch first", no form.
2. Create a batch on Site batches, return to VLB Manual. Expect: the batch auto-opens, location, date, shift and shift hours are fixed, Shift shows a Shift Master row.
3. Enter one card and save. Expect: card number VLB_M/<date>/<shift>/001, card listed in the batch register.
4. In another browser, as the batch owner, Lock VLB. Within ten seconds the first browser should replace the form with "VLB locked, ask the site PM to unlock". No 409 at save.
5. Unlock VLB with the PM OTP, return to VLB Manual. Expect: form available again, Edit enabled on the saved card.
6. Pick a site whose shift has no Shift Master row. Expect: red notice naming the shift and site, Save disabled.

**Script B: Scan and PDF upload, release 1**

1. VLB Scan for a site with no open batch. Expect: Scan setup shows "No open VLB batch for <site>, <date>, shift <n>", upload controls disabled.
2. Create the batch, return. Expect: upload enabled, context fixed.
3. Upload a ten-page PDF. Expect: ten cards queued, extraction proceeds, the context gate holds only if a card disagrees on date, site or shift.
4. Try to approve a card while another card is mismatched. Expect: refused with the gate message. Discard the mismatched card, approve again. Expect: approved.
5. Lock VLB from another browser, then Save all. Expect: refused with the VLB locked message, no cards posted. Unlock, Save all again. Expect: all posted with VLB_S numbers.
6. Upload the same PDF again through the Windows scanner "PC files" path (release 3). Expect: refused as already uploaded.

**Script C: Scan posting rule, release 2**

1. Option A: upload three clean cards, do not approve. Expect: none posts. Approve one. Expect: it posts within a few seconds, the other two wait.
2. Option B: switch "Post complete cards automatically" off in Scan setup, upload three clean cards. Expect: none posts, banner absent. Switch on. Expect: banner shown, clean cards post.

**Script D: Form quality, release 3**

1. Enter closing KMR below opening. Expect: breach text beside the field, signed negative total; save blocked or warned per the decision taken.
2. Enter working HMR above shift hours. Expect: warning beside the field, save allowed, breach visible in Saved VLB.
3. Pick a loading machine from another site in a trip row. Expect: tagged in the dropdown, save succeeds, the batch holds panel lists the machine with the express transfer button.
4. Open a saved card in a locked batch. Expect: Edit disabled with reason, no Delete button, First and Last stay within the batch.

# 7. Deployment day

## 7.1 Timing

Deploying logs every user out within ten seconds, and an MIS user mid-card loses unsaved form fields. Deploy between shifts. Sites run up to three shifts a day, so the window after a batch has been closed and before the next shift's cards arrive is the right one; confirm with the site MIS users which hour that is on the day. Scan uploads in flight are safe: cards already uploaded stay in the scan batch and extraction resumes when the server is back.

## 7.2 Sequence

1. Announce the window to MIS users and site PMs the day before, with the one-paragraph note from section 8.
2. Confirm no batch is mid-Save all: Site batches shows no scan entry in "saving" state at the chosen time.
3. Merge the pull request to main. The deploy workflow starts on its own.
4. Watch the workflow: install, OCR prepare, build, test, release identity, server start, deploy, health poll. It fails closed at any red step and the live site is untouched.
5. When the health poll passes, open pulse-ll.cmll.in/api/health and confirm status ok, database connected and the new commit.
6. Run Script A steps 1 to 3 and Script B steps 1 to 2 on the live site with a test batch at one site, then delete that batch through the Deletion page if it was not a real shift.
7. Tell the users the window is over.

```
Invoke-WebRequest https://pulse-ll.cmll.in/api/health | Select-Object -ExpandProperty
Content
```

## 7.3 Environment settings

| Setting | Value | Note |
|---|---|---|
| VLB_TWO_STAGE_BATCHES | unset (on) | Unchanged by these releases. Change 1, 2, 7 and 8 read it and fall back to single-step behaviour when it is 0. |
| LOCAL_SCAN_ENABLED | as today | Change 2 and 9 touch the local scanner routes; no new setting. |
| GEMINI_API_KEY | set by the workflow | Unchanged. Change 9 alters render size for the local path only; verify extraction on real cards before merging R3. |

## 8. Note to users

Send this to MIS users and site PMs with release 1. Adjust the time.

```
Pulse-II update on <date> at <time>, between shifts. You will be logged out once and
can sign in again straight away.

What changes:
- VLB Manual and VLB Scan only open a batch you can actually write to. A locked or
closed batch shows the reason instead of a form.
- Scanning needs the VLB batch to exist first. Create it on Site batches, then scan or
upload.
- If a shift has no Shift Master row for your site, the form tells you before you type
anything.
- Approve waits until wrong-date or wrong-site cards in the entry are removed.

Nothing about saved cards, batch numbers or reports changes. Problems: message the MIS
admin group with the site, batch number and a screenshot.
```

## 9. Rollback

- Every release is code only (release 2 option B adds one nullable column that an old build ignores). Rollback is a revert on main, which redeploys through the same workflow in about fifteen minutes: git revert <merge commit>, push.
- Reverting release 1 reopens the gaps the review found but loses no data, because the server rules it depends on predate it.
- Reverting release 2 option A resumes automatic posting of clean cards; cards left unapproved during the release post as soon as the old build is live, so tell reviewers before reverting.
- Reverting release 3 restores the old render size for the local scanner; cards already extracted are unaffected.
- The two-stage flag VLB_TWO_STAGE_BATCHES=0 remains the rollback for the batch stage model itself and is independent of these releases.

```
git revert -m 1 <merge-commit-sha>
git push origin main
```

## 10. Decisions needed before building

| Decision | Options | Needed for | Recommendation |
|---|---|---|---|
| Scan posting rule | A: approval required for every card. B: auto-post stays, per-batch switch, banner, docs corrected. | Release 2 | A if the team treats the Approve click as the control point; B if shift throughput matters more. Either way, code and documentation must agree. |
| Closing below opening | Block the save, or warn and store as today. | Release 3, change 6 | Block. A negative reading is always a transcription error. |
| Who may add cards to another user's open batch | Keep as today (any site-authorised MIS user), or owner only. | Release 1, change 1 wording | Keep as today; document it in PROJECT_WORKFLOW.md. |
| Deployment window | Hour between shifts per site. | Every release | Ask the site MIS users; the same hour works for all three releases. |

## 11. Effort and order

| Release | Changes | Build and test | Local walk-through | Live verification |
|---|---|---|---|---|
| R1 | 1, 2, 3, 4 plus the CRLF test fix | 2 to 3 working days | Half a day with an MIS and a PM login | Scripts A and B, one site, thirty minutes |
| R2 | 5A or 5B | Half a day (A) or 1 day (B) | One hour | Script C, one site |
| R3 | 6, 7, 8, 9 | 3 to 4 working days | Half a day, plus ten real cards through the local scanner for the render size check | Script D, one site, then watch the next full shift at all sites |

Release 1 first, at least one full shift cycle at every site on release 1 before release 2, and release 3 after the team has given the two decisions in section 10.